Changelog

All significant changes to the SHA-256 + CIE Protocol are documented here.
For full critical analysis of each issue identified and resolved, see Appendix A of the paper.

[1.0.0] — 2026 (this release)

Added

- **Part 8.0** — Hash of the print photograph: each physical copy is now cryptographically anchored to the certificate via a signed photograph taken in the same CIE session
- **Part 13** — Transfer of Ownership: a Transfer Certificate creates a mathematically verifiable provenance chain across all subsequent sales, without any platform
- **Section 13.3b** — Guidance for non-Italian and non-EU buyers
- **Part 12.3** — Offline HTML verifier for collectors: green/red light, zero technical expertise required
- **Part 12.1** — Complete explanation of SHA-256, SHA-512, and BLAKE2b with clear hierarchy (SHA-256 mandatory, BLAKE2b optional additive only)
- **Pilot case** — Il Ticket Infinito (March 2026): first real documented application, with transparent declaration of gaps
- English translation of full paper and all tools

Changed

- FEA vs QES distinction moved from Appendix to the heart of Layer 4, before signing
- PEC now explicitly described as excellent practical redundancy, not a qualified TSA
- “Private deed with enhanced probative value under Legislative Decree 82/2005 (CAD) and EU Regulation 910/2014 (eIDAS)” replaces the imprecise “authenticated private deed”
- Section 8.3 generalised from photography-specific to all artistic media
- DOI explained for non-academic readers; corpus reference made optional
- “90% of transactions undocumented” replaced with honest qualitative observation (the phenomenon is structurally unmeasurable)
- Assertive tones softened throughout

Fixed

- Double “pursuant to” in legal clause (Part 1.2 and edition clause)
- Missing FES (Simple Electronic Signature) disclosure for scanned handwritten signatures in transfer procedure
- BLAKE2b incorrectly presented as primary recommendation; repositioned as optional

Tools

- `sha_generator.html` / `sha_generator_en.html` — offline multi-algorithm hash calculator
- `sha_verifier.html` / `sha_verifier_en.html` — offline certificate verifier with green/red light

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Previous development versions

The protocol underwent iterative refinement through internal study versions before this public release. The following issues were identified and resolved across those versions:

Issue	Problem	Resolution
1	Firma4NG implied usable on smartphone	Explicit distinction Firma4NG (PC) vs CieSign (mobile)
2	CIE timestamp described as Ministry timestamp	PEC added as mandatory Layer 5
3	AES/FEA equated to authenticated deed	AES vs QES distinction, burden of proof explained
4	SHA Document not in hash chain	Advanced level documented in Part 8
5	Zenodo dependency not acknowledged	Contradiction declared; personal backup recommended
6	CIE expiry not addressed	Section 11.4 + .p7s report recommendation
7	No physical-digital link per copy	Part 8.0 — hashed print photograph
8	No ownership transfer covered	Part 13 — Transfer Certificate
9	PEC presented as equivalent to TSA	Clarified as practical redundancy; TSA option added
10	“Authenticated private deed” terminology	Replaced with precise legal formulation
11	Photo verification misunderstanding possible	Clarification box: photo is archival anchor, not collector tool
12	Adobe Reader required for collector verification	Part 12.3 — offline HTML verifier

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